

Title of the Note

The one question this note answers?

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One sentence stating the subject and that it is developed from first principles.

Prerequisites. The mathematics the reader is assumed to have, listed concretely. **Nothing beyond that is assumed.** Name any advanced machinery the notes develop for themselves rather than assume.

What you should be able to do afterwards. Three or four concrete capabilities, phrased as verbs: derive this result rather than recall it, state exactly what that theorem does and does not assert, apply the method to the case study.

Companion notes. How this note relates to its siblings in the series, and which sections carry the connections (for example §2.2).

Reading map

Part 0 — Vocabulary. §0 fixes notation and states every result used but not derived. Skip it and refer back as needed.

Part I — Foundations. §1 sets up the object of study; §2 derives the main result.

Part II — Practice. §3 is the worked case; §4 states the limits. Appendix A records the numerical checks.

Part 0 — Notation and vocabulary

0. Everything used but not derived

These notes are self-contained. §0.1 fixes notation; §0.2 states the results used, with a proof where one is short and an explicit statement of trust where it is not. Nothing later uses a symbol, result or term that does not appear here or at its point of first use.

0.1 Notation

Symbol	Meaning
x, y	placeholder: the primary variables
θ	placeholder: the parameter being estimated
$f(x; \theta)$	placeholder: the model, with a cross-reference to where it is defined (Definition 0.1)
$\mathbb{E}, \text{Var}$	expectation, variance
$\log$	natural logarithm throughout
■	end of proof

0.2 Mathematical results used

Definition 0.1 (name of the concept). Placeholder definition. The term being defined is set in **bold** at its point of definition; the rest of the statement is upright text with inline mathematics such as $f(x; \theta)$.

Lemma 0.2 (name of the lemma). Placeholder statement of a short auxiliary result, e.g. if $a \leq b$ then $\varphi(a) \leq \varphi(b)$ for every non-decreasing φ .

Proof. Placeholder proof, ending with the filled square the class supplies automatically. ■

Fact 0.3 (name of the fact). Placeholder statement of a standard result used without proof. Say where a proof can be found [2]; a Fact is the only statement kind permitted to appear without one.

Part I — Foundations

1. First topic

Placeholder paragraph motivating the section: what problem it solves and why the obvious approach fails. Display equations pad their top-level relations with thick spaces:

$$y = f(x; \theta) + \varepsilon, \quad \mathbb{E}[\varepsilon] = 0.$$

Proposition 1.1 (name of the proposition). Placeholder statement of an intermediate result.

Proof. Placeholder proof, citing Lemma 0.2 and Fact 0.3 by name rather than by bare number. ■

Placeholder plain-language restatement: two or three sentences saying what the derivation above established, without symbols where possible.

2. Second topic

2.1 First subsection

Theorem 2.1 (name of the main result, Author [1]). Placeholder statement of the main theorem. An equation that is referred to later is numbered and labelled:

$$F(\theta) = \mathbb{E}[\ell(y, f(x; \theta))]. \tag{2.1}$$

Proof. Placeholder proof. Show every intermediate step; cite Eq. (2.1) by reference. ■

2.2 Second subsection

Placeholder derivation leading to the section's final result, which is framed:

$$\theta^* = \arg \min_{\theta} F(\theta).$$

Corollary 2.2 (name of the corollary). Placeholder consequence that follows from Theorem 2.1 in a line or two.

The results combine into a procedure (Figure 1 illustrates it).

Algorithm 2.3 (name of the procedure).

1. Placeholder first step.
2. Placeholder second step, citing the result it relies on (Theorem 2.1).
3. Placeholder final step.



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\includegraphics{figures/name.png}
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Figure 1: Placeholder caption, which is the figure’s interpretation. Its first sentence names the subject or states the finding; then say what the axes are, what each panel shows (**(a)**, **(b)**), and what pattern to look for. **The takeaway:** the single conclusion the reader should draw. *Generated by `scripts/plot_name.py`.*

Part II — Practice

3. Case study

Placeholder worked case: the method applied to a concrete instance, with the measured numbers and what they mean. **State the lesson once, in bold, and generally.**

4. Assumptions and where they fail

Placeholder list of the assumptions the derivation made, and for each the circumstances in which it breaks and what then goes wrong.

Appendix A: Numerical verification

Every analytic claim above was checked numerically; seeds are fixed so the checks reproduce.

Claim	§	Check	Result
Placeholder claim	2	how it was checked	measured agreement
Placeholder claim	2.2	how it was checked	measured agreement

References

[1] Surname, A. (Year). *Title of the article*. Journal, Volume(Issue), pages. [Link](#)

[2] Surname, A., & Surname, B. (Year). *Title of the Book*, edition, chapter. Publisher. [Link](#)